

RESEARCH

Open Access



Integrating CT radiomics and clinical features using machine learning to predict post-COVID pulmonary fibrosis

Qianqian Zhao^{1,2,3†}, Yijie Li^{1,2,3†}, Chunliu Zhao^{4†}, Ran Dong⁵, Jiaxin Tian^{1,2,3}, Ze Zhang⁶, Lin Huang^{1,2,3}, Jingwen Huang^{1,2,3}, Junhai Yan⁴, Zhitao Yang⁷, Jiangnan Ruan⁸, Ping Wang⁸, Li Yu^{5*}, Jieming Qu^{1,2,3*} and Min Zhou^{1,2,3*}

Abstract

Background The lack of reliable biomarkers for the early detection and risk stratification of post-COVID-19 pulmonary fibrosis (PCPF) underscores the urgency advanced predictive tools. This study aimed to develop a machine learning-based predictive model integrating quantitative CT (qCT) radiomics and clinical features to assess the risk of lung fibrosis in COVID-19 patients.

Methods A total of 204 patients with confirmed COVID-19 pneumonia were included in the study. Of these, 93 patients were assigned to the development cohort (74 for training and 19 for internal validation), while 111 patients from three independent hospitals constituted the external validation cohort. Chest CT images were analyzed using qCT software. Clinical data and laboratory parameters were obtained from electronic health records. Least absolute shrinkage and selection operator (LASSO) regression with 5-fold cross-validation was used to select the most predictive features. Twelve machine learning algorithms were independently trained. Their performances were evaluated by receiver operating characteristic (ROC) curves, area under the curve (AUC) values, sensitivity, and specificity.

Results Seventy-eight features were extracted and reduced to ten features for model development. These included two qCT radiomics signatures: (1) *whole lung_reticulation (%) interstitial lung disease (ILD) texture analysis*, (2) *interstitial lung abnormality (ILA)_Num of lung zones $\geq$ 5%_whole lung_ILA*. Among 12 machine learning algorithms evaluated, the support vector machine (SVM) model demonstrated the best predictive performance, with AUCs of 0.836 (95%

[†]Qianqian Zhao, Yijie Li and Chunliu Zhao contributed equally to this work.

*Correspondence:

Li Yu
96778@tongji.edu.cn
Jieming Qu
mq0906@163.com
Min Zhou
doctor_zhou_99@163.com

Full list of author information is available at the end of the article



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

CI: 0.830–0.842) in the training cohort, 0.796 (95% CI: 0.777–0.816) in the internal validation cohort, and 0.797 (95% CI: 0.691–0.873) in the external validation cohort.

Conclusions The integration of CT radiomics, clinical and laboratory variables using machine learning provides a robust tool for predicting pulmonary fibrosis progression in COVID-19 patients, facilitating early risk assessment and intervention.

Keywords Post-acute COVID-19 sequelae, Pulmonary fibrosis, Quantitative CT, Radiomics, Machine learning

Introduction

The long-term multiorgan system consequences of COVID-19 deserve the attention of researchers. Symptoms can last for months or even years in a subset of COVID-19 patients and are defined as post-COVID syndrome, long COVID-19, or post-acute sequelae of COVID-19 (PASC) [1, 2]. One study summarized the proportion of each symptom change over time during a 2-year follow-up study in China [3]. Studies have reported persistent respiratory sequelae as predominant features, with dyspnea and cough constituting the most prevalent prolonged pulmonary symptoms [4, 5]. Notably, 6-month follow-up chest CT imaging revealed pulmonary fibrosis (PF) in approximately one-third of COVID-19 survivors [6]. Post-COVID-19 pulmonary fibrosis (PCPF) portends substantial clinical burdens, including elevated mortality risks, restrictive ventilatory dysfunction, diminished exercise tolerance, and impaired quality of life [7, 8].

Computed tomography (CT) has solidified its role as an indispensable diagnostic tool in diagnosing diseases, with its applications continually expanding. As the reference standard for interstitial lung disease (ILD) and pulmonary nodule evaluation, its diagnostic value is further amplified through quantitative analysis (qCT). This technique can be used to evaluate the percentage of lungs affected by the disease through automatic, reproducible, and quantifiable metrics. Compared with conventional image review, qCT enables better characterization of disease phenotypes and deepens our understanding of variable patient prognoses. Radiomics is a high-throughput imaging feature extraction paradigm [9, 10] that has improved diagnostic precision in lung disease assessment by using chest CT data [11]. Machine learning algorithms excel in identifying complex patterns within high-dimensional radiomic data, enabling the development of robust predictive models that surpass traditional statistical methods [12, 13]. Emerging evidence highlights the potential of qCT in COVID-19 management, with standardized metrics proposed for disease quantification [14]. These indicators can be used for PCPF prediction using various machine learning algorithms.

While preliminary studies have delineated the pulmonary CT manifestations of COVID-19 during the acute and post-acute phases [15, 16], current follow-up

assessments predominantly rely on subjective visual grading systems [17, 18]. Although early qCT applications demonstrate prognostic potential [19, 20], comprehensive longitudinal studies employing whole-lung radiomic analysis remain imperative to establish PF prevalence and progression patterns. This knowledge gap underscores the critical need for validated quantitative tools enabling early PCPF risk stratification and effective disease management. To address this unmet clinical demand, we propose constructing an integrative prediction model that integrates qCT biomarkers with clinical indicators for PCPF risk identification, employing machine learning algorithms to train and validate the model.

Materials and methods

Study design and patients

This multicenter retrospective study included a total of 1251 adult patients with confirmed COVID-19 pneumonia from three hospitals (Ruijin Hospital, Tongji Hospital and Ruijin Hospital, LuWan Branch) in Shanghai from December 2022 to October 2023. This study was approved by the ethics committees of all participating centers (No. 2024-73) and was performed according to the principles of the Declaration of Helsinki. Because of its retrospective nature, the need to obtain informed consent from the participants was waived. Patients were screened according to the following inclusion criteria: (1) aged ≥ 18 years; (2) SARS-CoV-2 infection confirmed by positive quantitative reverse-transcription polymerase chain reaction (qRT-PCR) testing according to World Health Organization's interim guidance [21]; (3) a diagnosis of COVID-19 pneumonia by chest CT scan; and (4) received at least one thin-section chest CT scan one month after being diagnosed with COVID-19. For patients with multiple eligible CT scans, the most recent examination was chosen. The exclusion criteria were as follows: (1) complicated with any autoimmune disease; (2) had lung fibrosis before being infected with COVID-19; and (3) did not undergo two chest CT scans. After applying these criteria, 204 patients were included in the final analysis.

The diagnosis and clinical severity classification of pneumonia were made according to the Diagnosis and Treatment Protocol for COVID-19 Patients (10th

edition) issued by the National Health Commission of the People's Republic of China [22]. Ninety-three patients were included in the model development cohort and randomly divided into training and internal validation sets at an 8:2 ratio. In addition, 111 patients from three hospitals were assigned to the external validation cohort. The clinical and laboratory parameters, including demographic characteristics, comorbidities and medication status, were collected from electronic medical records at admission. Some items with missing values greater than 30% were excluded from subsequent analyses.

CT acquisition technique and lung fibrosis assessment

All participants underwent non-contrast chest CT scans while breath-holding and lying in a supine position with arms above head. The CT acquisition parameters are shown in Table 1. The CT images obtained during follow-up, along with complete electronic health records, were independently and comprehensively evaluated by two respiratory specialists (with 23 and 12 years of experience, respectively). The pulmonary sequelae of COVID-19 patients were classified into those with residual pulmonary fibrotic changes (PF group) and those without fibrosis (non-PF group). In cases where the opinions of the two respiratory experts were discordant, a third expert with 30 years of experience was consulted for further clarification.

Whole lung segmentation and CT image analysis

All chest CT images were processed using AVIEW software (version 1.1.46.65; Hangzhou Shimai Intelligent Technology Co. Ltd., China) for whole lung segmentation. The software employs a modified U-net deep learning model to automatically segment the whole-lung images of all chest CT scans and classify patterns for ILD [23, 24]. Both lungs were divided into six lung zones following the standard anatomic landmarks (inferior aortic arch and right inferior pulmonary vein) (Fig. 1). The regional CT patterns of lung parenchyma include: (1) normal, (2) consolidation, (3) reticulation, (4) ground-glass opacity (GGO), (5) emphysema, and (6) honeycombing, which were quantified using the deep learning algorithm as described in the reference [25]. Also, a previous study has validated the performance of this program [26].

Radiomic feature extraction

Radiomic features are quantitative image features extracted in a high-throughput manner from medical images. In our study, they were automatically extracted from pulmonary regions of interest (ROI) using AVIEW software for qCT analysis. The specific methods used are as follows.

Attenuation area analysis: CT voxels represent three-dimensional units of X-ray attenuation coefficients (measured in Hounsfield units, HUs) within the pulmonary parenchyma. High attenuation area (HAA) was quantified by determining the percentage of the imaged lung volume that exhibited attenuation values within the range of −600 to −250 HU [27, 28]. Low-attenuation area (LAA) was obtained below a threshold of −950 HU that was regarded as emphysema. Besides, the texture and shape analysis might be influenced by the major pulmonary vessels or chest wall of the lung parenchyma. Therefore, areas with HU between −400 and −1024 were segmented for clipping major pulmonary vessels by thresholding control during images pre-processing [29]. Last, lung attenuation was measured and recorded for each lobe on baseline chest CT images of all participants.

Interstitial lung abnormality (ILA) analysis: ILA is described as incidental CT findings of non-dependent abnormalities affecting more than 5% of any lung zone at complete or partial chest CT where ILD was not previously suspected [30]. All images were divided into two categories: fibrotic ILA (including architectural distortion and traction bronchiectasis) and non-fibrotic ILA (the sum of GGO and reticulation) according to the results of automated identification and quantification based on texture-based analysis and deep learning approaches (Fig. 1B). The total ILA was defined as the sum of fibrotic ILA and nonfibrotic ILA values [26]. The software identified ILA on chest CT when the maximum extent of the total ILA was at least 5% in any of the six zones.

ILD texture analysis: ILD is comprised of heterogeneous disorders characterized by inflammation and/or fibrosis of the pulmonary interstitium, leading to progressive lung function decline. We performed whole-lung ILD segmentation on the original CT slices using the AVIEW software and divided the selected region of interest (ROI) into six types of image patterns as mentioned above. The process of quantifying the spatial distribution of the disease was described in detail in the previous study [25]. The regional fractions and volumes of ILD

Table 1 CT acquisition parameters for different scanners

Device name	Matrix	Pitch	Slice thickness (mm)	The field of view (mm ²)	Tube voltage (kV)	Tube current (mAs)
Siemens Drive	512*512	0.992	1	350*350	120	SemiCAREkv
GE Revolution	512*512	1.2	1-1.25	350*350	120	SmartmA
Aquilion ONE Pure Vision	512*512	1.0875	1	350*350	120	50

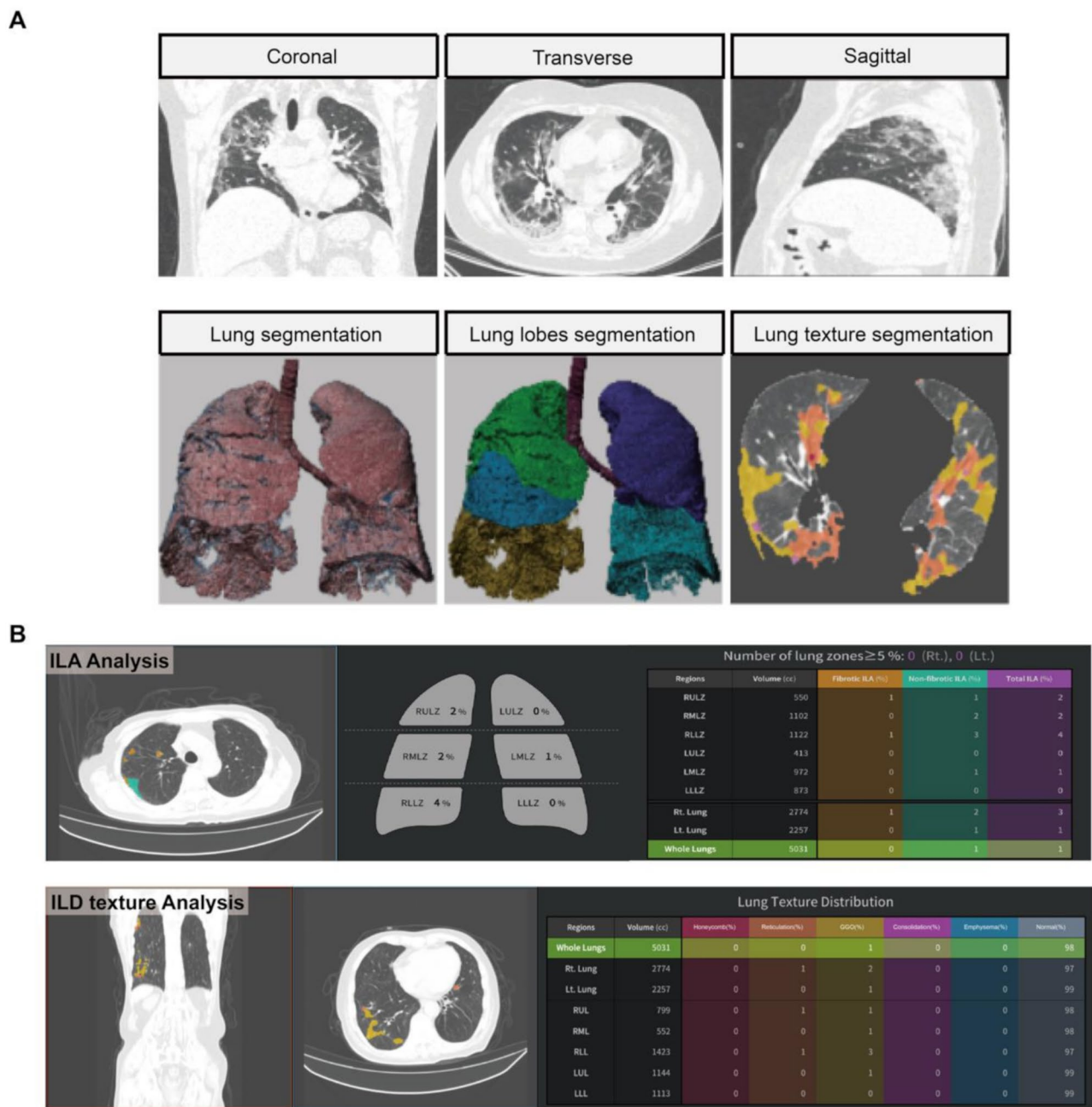


Fig. 1 Demonstration of the main process for automated lung segmentation and quantitative analysis by utilizing the AVIEW software. **(A)** Segmentation results of different lung regions on the transverse, coronal, and sagittal planes. **(B)** Example of ILA analysis. **(C)** Example of ILD texture analysis. Different colors represent different regions and characteristics. ILA, Interstitial lung abnormalities; ILD, Interstitial lung disease; GGO, ground-glass opacity

disease patterns in pixel-level serve as radiomics features for each CT scan pattern. A total 12 features calculated automatically were extracted as the radiomics features related to ILD (Fig. 1C and Table S2 in the Supplementary Materials).

Feature selection and model construction

Z score normalization was applied to all features to eliminate scale differences. For feature selection, a least

absolute shrinkage and selection operator (LASSO) model was applied by incorporating an L_1 regularization term into the loss function. This convex optimization approach induced sparsity in the coefficient vector β , systematically shrinking insignificant feature weights toward zero while preserving discriminative predictors. Five-fold cross-validation was conducted to select the penalty term lambda (λ). Twelve different machine learning algorithms (Table S1 in the Supplementary Materials)

were rigorously evaluated by precision–recall curves and receiver operating characteristic (ROC) analysis. The optimal model was selected based on the maximal area under the ROC curve (AUC-ROC) with simultaneous consideration of computational efficiency.

Statistical analysis

In this study, Python 3.9 (<https://www.python.org/>) and R4.3.2 (<https://www.Rproject.org>) were used for graphic production and statistical analysis. Continuous variables are presented as the mean±standard deviation (SD) or the median and interquartile range (IQR), as appropriate. Categorical variables are presented as numbers (%). Student’s unpaired t test was used to compare normally distributed continuous variables between groups. Conversely, for data that did not adhere to a normal distribution, we utilized the Mann–Whitney U test. Categorical variables were compared by the χ^2 test or Fisher’s exact test. The “tableone” R package (v0.13.2) facilitated a comprehensive comparison of clinical and demographic

characteristics across cohorts. LASSO regression with 5-fold cross-validation was conducted using the “sklearn.linear_model.LassoCV” Python package, optimizing λ through the “lambda-se” heuristic. Model performance evaluation utilized sklearn.metrics modules. ROC curves were generated using roc_curve, with AUC quantified via roc_auc_score. *P*value<0.05 was considered to indicate statistical significance.

Results

Patient characteristics

This study included 204 confirmed COVID-19 pneumonia patients admitted to Ruijin Hospital, Tongji Hospital and Ruijin Hospital LuWan Branch. The workflow of this study is shown in Fig. 2. The baseline characteristics of the patients in the development and validation cohorts are presented in Table 2. Disease severity stratification revealed 100 moderate cases and 104 severe or critical cases. Based on follow-up chest CT scans and clinical records evaluated by respiratory specialists, 103 patients

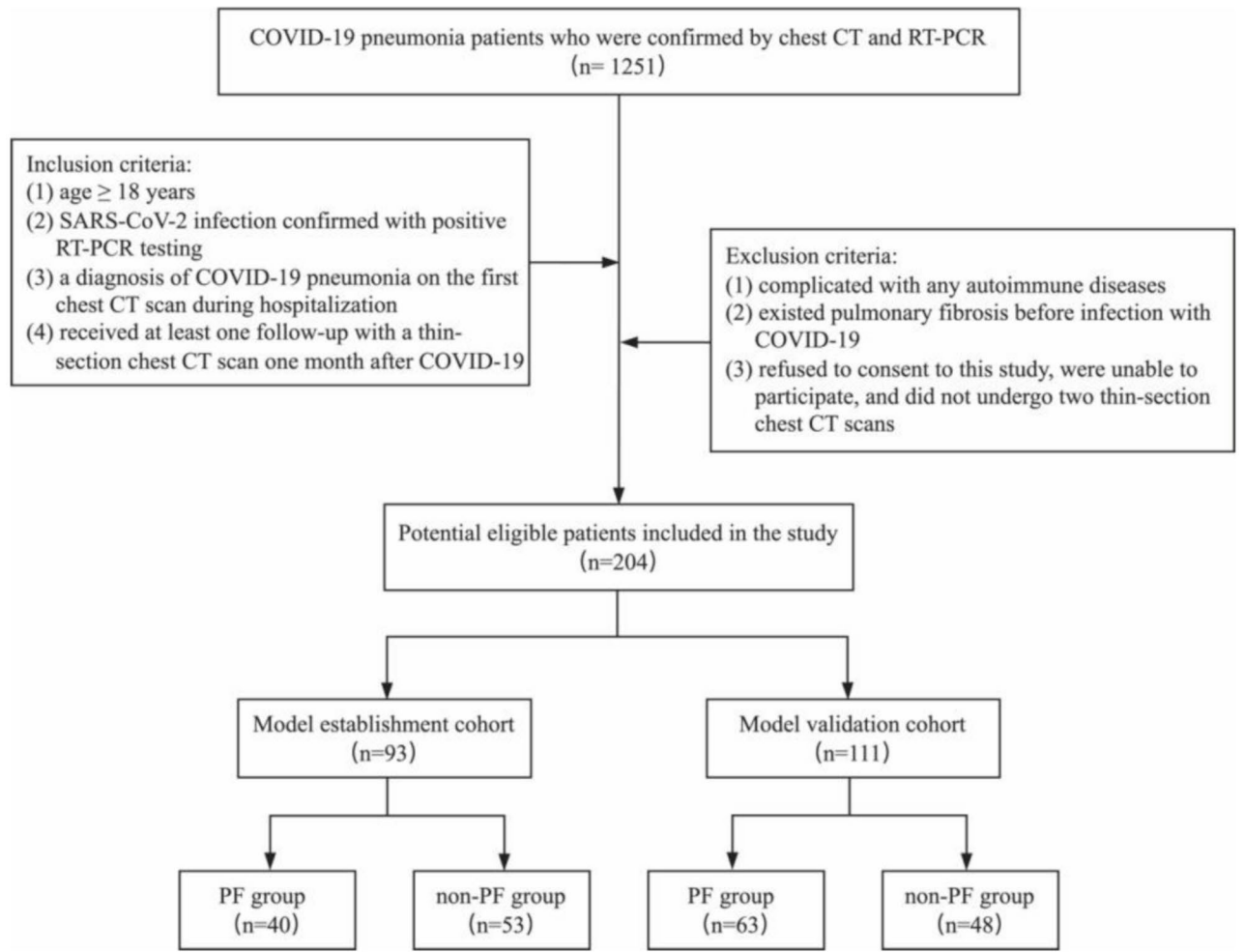


Fig. 2 Enrollment flowchart of the study

Table 2 Baseline characteristics of the study cohort

Clinical features		Model development cohort (n = 93)			External validation cohort (n = 111)		
		PF group (n = 40)	non-PF group (n = 53)	p value	PF group (n = 63)	non-PF group (n = 48)	p value
Socio-demographic details							
Age(years)		74.8 ± 9.77	67.1 ± 15.2	0.004	75.6 ± 10.7	69.5 ± 9.93	0.003
Male sex		30 (75.0)	34 (64.15)	0.263	42 (66.67)	30 (62.5)	0.649
BMI		23.8 ± 3.2	24.3 ± 4.13	0.563	23.9 ± 3.75	23.7 ± 4.0	0.755
COVID-19 pneumonia severity classification	moderate	11 (27.5)	32 (60.38)	0.003	23 (36.51)	34 (70.83)	0.001
	severe	16 (40.0)	15 (28.3)		34 (53.97)	12 (25.0)	
	critical	13 (32.5)	6 (11.32)		6 (9.52)	2 (4.17)	
Clinical characteristics							
Neutrophil (10 ⁹ /L)		5.82 (3.75, 7.94)	4.71 (3.11, 6.30)	0.017	5.23 (3.33, 8.04)	3.91 (2.54, 5.57)	0.037
NLR		9.25 (6.57, 16.86)	5.39 (3.47, 9.42)	0.001	6.50 (3.76, 9.30)	4.16 (2.43, 8.07)	0.080
Neutrophil (%)		85.75 (79.52, 89.93)	76.7 (70.5, 84.3)	0.001	80.0 (69.65, 85.45)	73.4 (63.08, 82.5)	0.145
Lymphocyte (%)		9.25 (5.22, 12.72)	14.3 (8.8, 19.5)	0.001	11.8 (8.75, 18.65)	17.25 (10.63, 25.75)	0.073
LDH (IU/L)		291 (228.75, 351.25)	246 (198, 326)	0.090	300.2 (227.07, 350.5)	233.5 (190.25, 270)	0.001
Myoglobin (ng/mL)		25.85 (10.65, 57.6)	26.5 (15.8, 55.1)	0.898	54 (29.75, 98.75)	33.3 (19.4, 62.05)	0.007
CK-MB (ng/mL)		1.7 (1.17, 14.47)	1.2 (0.7, 2.7)	0.020	2.1 (1.3, 3.1)	1 (0.8, 1.5)	<0.001
D-dimer (pg/mL)		0.95 (0.53, 2.06)	0.56 (0.34, 1.31)	0.034	0.64 (0.34, 1.02)	0.34 (0.26, 0.72)	0.002
NT-proBNP (pg/mL)		397.9 (211.6, 881.8)	193.8 (112.3, 461.75)	0.041	125.5 (82.15, 243.55)	120.3 (60.85, 180.35)	0.425
Clinical treatment							
Anticoagulants		36 (90.0)	36 (67.9)	0.023	59 (93.7)	37 (77.1)	0.024
Anti-fibrotic drugs		16 (40.0)	7 (13.2)	0.006	10 (15.9)	0 (0.0)	0.010
Follow-up interval		186.15 ± 131.63	266.06 ± 144.24	0.007	222.40 ± 157.36	248.92 ± 139.05	0.357

Data are presented as mean ± SD or median (IQR). BMI, Body mass index; NLR, Neutrophil-to-lymphocyte ratio; NT-proBNP, N-terminal pro-brain natriuretic peptide

were diagnosed with PF, and 101 patients were classified as non-PF. Owing to the lack of follow-up CT images for most nonfibrotic patients, they were excluded from the study, leading to a smaller cohort size than that of fibrotic patients. The development cohort comprised 40 PF patients and 53 non-PF patients, whereas the external validation set consisted of 63 PF patients and 48 non-PF patients. All patients were followed up for more than 30 days after disease onset to observe the continuous pulmonary consequences caused by COVID-19. In both the model development cohort and the external validation cohort, the PF group was significantly older and had more severe COVID-19 pneumonia than the non-PF group ($p = 0.004$ and $p = 0.003$; $p = 0.003$ and $p = 0.001$, respectively). Compared with non-PF patients, PF patients had significantly greater neutrophil counts, creatine kinase isoenzyme (CK-MB) levels, and D-dimer levels at admission. Importantly, PF patients in the entire cohort received anticoagulant therapy and antifibrotic drugs more frequently than non-PF patients.

Extraction of the radiomics signatures and feature selection

Using AVIEW software, we performed multi-level lung segmentation and conducted ILA and ILD texture analysis, automatically generating quantitative results after image import (Fig. 1). After extraction from the chest

CT images of those patients, 29 radiomics features were selectively retained for further analysis (Table S2 in the Supplementary Materials). We included laboratory examination findings, comorbidity information, demographic data, clinical medication and radiomics signatures in the model. Finally, the dataset of each patient contained a total of 78 features. To address high dimensionality and potential multicollinearity, we applied LASSO regression with 5-fold cross-validation for feature screening to choose the optimized subset of features. This process identified the optimal regularization parameter λ that retained 10 predictive features (Fig. 3A and B). The final feature set included age, neutrophil count, percentage of neutrophils, percentage of basophils, diabetes status, COVID-19 pneumonia severity classification, anti-fibrotic therapy, anticoagulant use, *whole lung reticulation (%) ILD texture analysis*, and *ILA_Num of Lung Zones ≥ 5%_Whole Lung ILA*. These features and their corresponding coefficients are shown in Fig. 3C. The λ values of the LASSO regression are shown in Table S3 in the Supplementary Materials.

Development and validation of a combined model

In this study, we implemented grid search with 10-fold cross-validation for hyperparameter optimization of each model. A predefined hyperparameter grid was established for every algorithm. Exhaustive search across

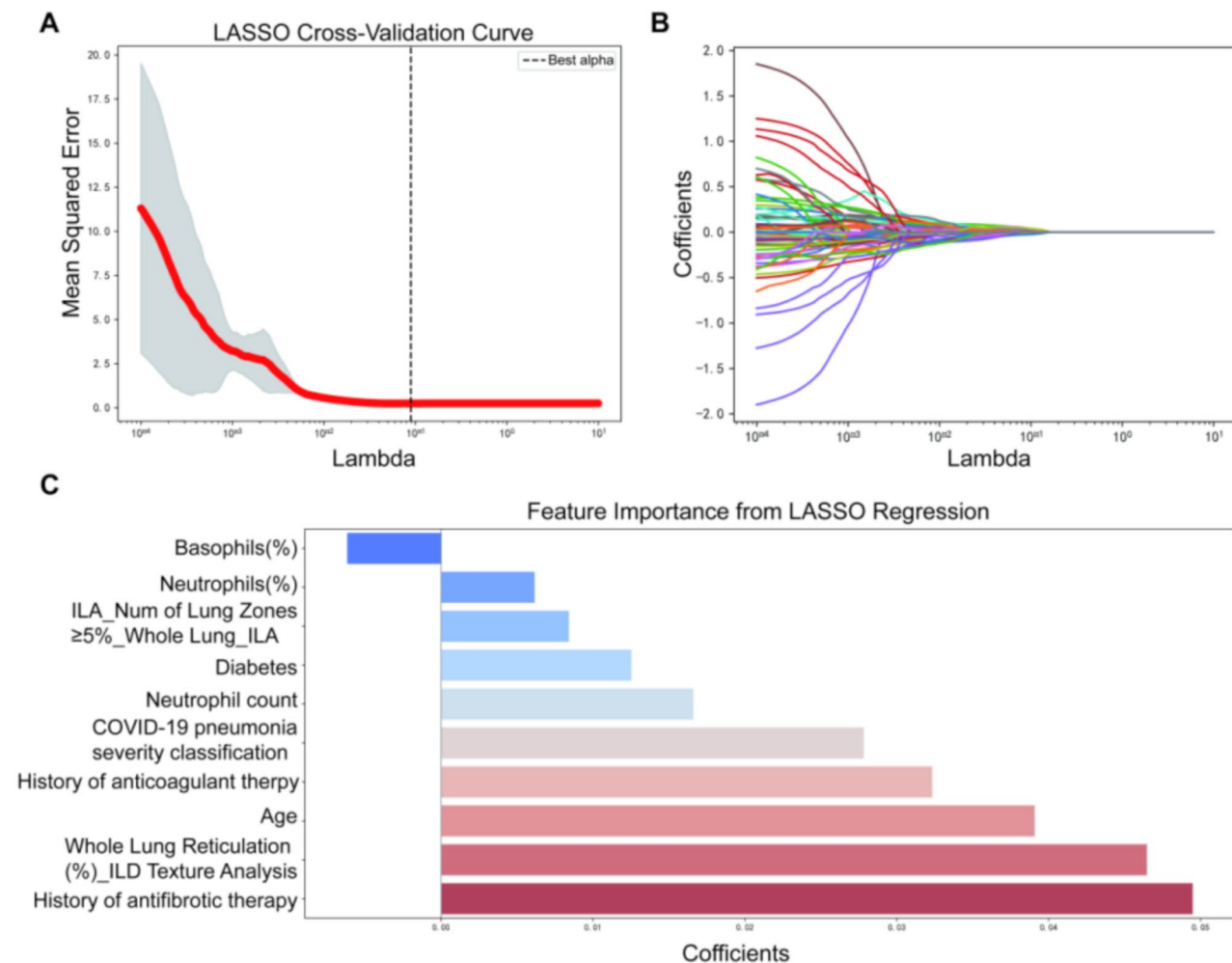


Fig. 3 LASSO regression of the radiomics signatures and clinical factors related to lung fibrosis. **(A)** LASSO cross-validation curve with the selected optimal hyperparameter λ values minimizing the mean squared error. **(B)** LASSO regression models identified the selected features by assigning them nonzero coefficients. **(C)** Ten significant features screened out by LASSO regression analysis, along with their corresponding coefficients. LASSO, least absolute shrinkage and selection operator; ILA, interstitial lung abnormalities; ILD, interstitial lung disease

all parameter combinations was performed exclusively on the training set. Feature dimensionality reduction was conducted solely on the training set using LASSO regression to prevent overfitting. Finally, model evaluation was performed strictly on the independent external validation set. The training set, validation set, and external validation set were independently segmented before the experiment. Last, 12 widely used machine learning models were compared. The average diagnostic performance of each model based on 50 random experiments across three cohorts is summarized in Table S4 of the Supplementary Materials. After comprehensive evaluation, the SVM model demonstrated the highest predictive performance. The ROC curves of the three cohorts are presented in Fig. 4. The constructed combined model showed strong discriminatory power, with AUCs of 0.836 (95% CI: 0.830–0.842) in the training cohort, 0.796 (95% CI: 0.777–0.816) in the internal validation cohort, and

0.797 (95% CI: 0.691–0.873) in the external validation cohort.

For each COVID-19 patient, the values of ten predictors identified and weighted during model construction were input into the SVM algorithm. The SVM algorithm computes a raw score for each patient based on predicted values and the optimal hyperplane position. This raw score is post-processed to generate a calibrated probability value. This probability value, ranging from 0 to 1, was used to estimate the individualized risk of future pulmonary fibrosis development. The probability output provides clinicians with valuable insights to guide personalized treatment strategies and prognostic evaluations.

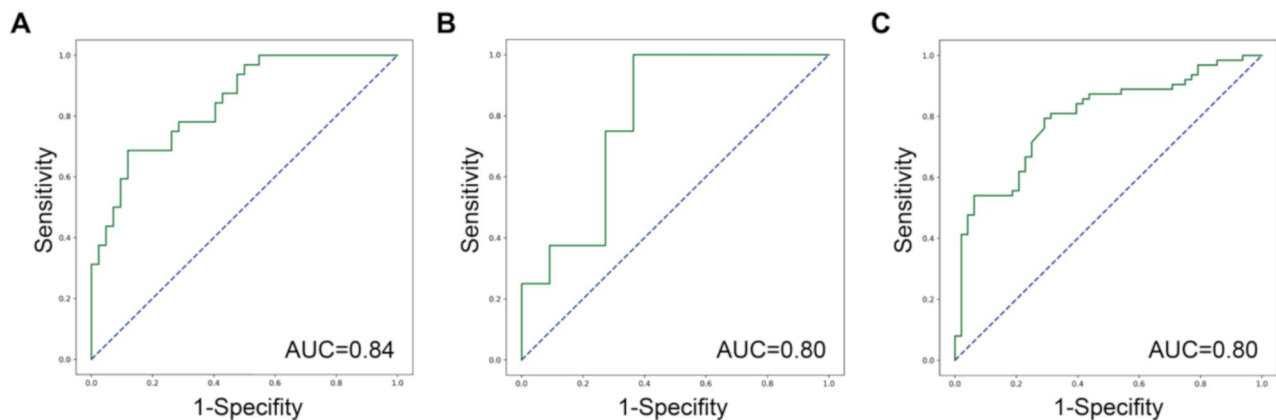


Fig. 4 ROC curves of the SVM model for predicting PF in COVID-19 pneumonia patients. **(A)** training cohort. **(B)** internal validation cohort and **(C)** external validation cohort. SVM, support vector machine; ROC, receiver operating characteristic

Discussion

Emerging evidence delineates a spectrum of COVID-19 sequelae [31, 32], with pulmonary fibrosis emerging as a predominant complication and a critical pulmonary manifestation of PASC [33]. Several studies have reported that 30–50% of COVID-19 survivors progress to lung fibrosis [34, 35]. This progression underscores a critical unmet need for early identification and intervention to halt fibrotic remodeling. However, clinical prediction tools for PCPF remain scarce and lack the ability to integrate multimodal biomarkers. In this study, we utilized quantitative analysis software that automatically detected lung lesions on chest CT scans and extracted key radiomic features. The results demonstrated that the radiomic system using machine learning performed well in predicting the risk of PCPF when combined with clinical data.

The outcome can be assessed through CT images at short-term (1–6 months) and long-term (12–24 months) follow-ups after recovery from COVID-19 [36]. A systematic review of 11 cohort studies indicated that CT-derived radiological features could predict the potential for pulmonary sequelae progression [37]. Damiano Caruso et al. developed a combined model for the prediction of 6-month fibrosis-like changes in COVID-19 patients [38], but their research assessed the 6-month outcomes in patients with pre-existing PF and did not perform a comparative analysis with COVID-19 patients without PF. Some studies have proposed factors affecting persistent PF in discharged COVID-19 patients [39, 40]; however, predictive models have not been established or have used only clinical parameters.

The final model incorporated 10 significant predictors, including two qCT parameters and eight clinical variables. Consistent with previous studies, older age remained a significant predictive factor for persistent PF [41–43]. Furthermore, the severity of COVID-19 was

associated with an increased risk of PF, where greater disease severity was correlated with greater fibrotic progression [44]. Anti-fibrotic therapy during hospitalization were identified as protective factors in previous study [45]. While antifibrotic use was a significant variable in the model (likely reflecting its association with disease severity), the analysis revealed that it was negatively correlated with a decrease in PCPF incidence or fibrosis resolution. This suggests its significance may derive from linkage to high-risk phenotypes rather than therapeutic efficacy. Crucially, our cohort included some patients with severe pneumonia at baseline. Clinically, for severe pneumonia patients with poor response to systemic glucocorticoids, nintedanib or pirfenidone is often added to prevent pulmonary fibrosis. However, these patients exhibit extensive lung involvement and inherent susceptibility to PCPF. Our data indicate that while anti-fibrotics may aid partial lesion resolution, they failed to prevent progressive fibrosis or achieve complete reversal in severe cases. Similarly, anticoagulant therapy is closely related to the patient's disease severity and coagulation profile. This underscores the need for future studies targeting PCPF regression and pulmonary rehabilitation efficacy in severe COVID-19. Elevated neutrophil counts and percentages are positively correlated with the risk of lung fibrosis, which aligns with prior reports that neutrophil counts are elevated in patients with severe COVID-19 and that individuals with PF changes at 3–6 months after recovery present upregulated neutrophil-associated immune signatures [46, 47]. In our study, diabetes mellitus as a comorbidity was significantly associated with chest CT abnormalities, which is consistent with prior evidence demonstrating that an uncontrolled glyce-mic status predisposed individuals to the development of PCPF [48]. In addition, our study revealed a negative correlation between an elevated basophil percentage and the risk of progressive PF, although the mechanistic role of basophils remains elusive. This association might

be supported by a prior study that indicated that BAL basophils at baseline were a novel risk factor for death in all ILD patients [49]. Our results suggest that this cell type and its role as a prognostic marker should be clarified in the future. Most importantly, we exploited two novel qCT biomarkers (*whole lung reticulation (%) ILD texture analysis* and *ILA_Num of lung zones $\geq 5\%$ whole lung ILA*), which exhibited good performance in predicting the risk of progressive fibrosis. To our knowledge, these indicators, especially the former, represent the first radiomic indices validated for the early prediction of COVID-19 patients at elevated risk of fibrotic lung remodeling, demonstrating superior discriminative capacity compared with traditional CT parameters.

However, there are several limitations to this study. First, this was a cross-sectional study, and it is necessary to conduct prospective research, including serial CT scans at 3-month intervals, to capture dynamic fibrosis progression in the future. Second, our model demonstrated validation robustness; however, the sample size of this study was relatively limited, and a larger cohort is still needed to confirm the generalizability of the model. In addition, the fact that our methodology was tested using 2D HRCT images must also be taken into account, as it may have caused the misclassification of lung parenchyma. In future, more accurate results could be obtained from the analysis of 3D images. Finally, retrospective data collection led to missing clinical variables in some patients, potentially introducing information bias and omitting relevant predictors that may influence fibrosis progression.

Conclusions

In conclusion, we developed and validated a clinical prediction model integrating radiomics CT features with machine learning to identify pulmonary sequelae, providing valuable guidance for early intervention and personalized management of COVID-19 patients at high risk of pulmonary fibrosis.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12931-025-03305-7>.

Supplementary Material 1

Author contributions

QZ, YL and CZ conceptualized and designed experiments, evaluated and interpreted data, and wrote the manuscript. JR and PW performed bioinformatics analysis. LY, JQ and MZ provided resources and guidance for experimental design and analysis. RD, JT, ZZ, LH, JH, JY and ZY provided guidance and provided resources. Min Zhou supervised the study. All authors reviewed and approved the final manuscript.

Funding

This work was supported by the National Key Research and Development Program of China (2024YFC3044400, 2024YFC3044600); Shanghai Targeted

Biomedical Emergency Project (23DX1900300); Noncommunicable Chronic Diseases-National Science and Technology Major Project (2023ZD0506200, 2023ZD0506201); The youth project of National Natural Science Foundation of China (82200004); Shanghai Jiao Tong University School of Medicine High Level Local University Innovation Team (SSMUZDCX20181001).

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study was approved by the Institutional Review Board of Ruijin Hospital, Shanghai Jiaotong University School of Medicine (Approval number: 2024-73). Information was collected from all participants after obtaining written or oral informed consent in accordance with the Declaration of Helsinki.

Consent for publication

Informed consents were obtained from all individual participants included in the study.

Competing interests

The authors declare no competing interests.

Clinical trial number

Not applicable.

Author details

¹Department of Pulmonary and Critical Care Medicine, Ruijin Hospital, Shanghai Jiao Tong University School of Medicine, No.197, Ruijin No.2 Road, Shanghai 200025, China

²Institute of Respiratory Diseases, Shanghai Jiao Tong University School of Medicine, No.197, Ruijin No.2 Road, Shanghai 200025, China

³Shanghai Key Laboratory of Emergency Prevention, Diagnosis and Treatment of Respiratory Infectious Diseases, Shanghai 200025, China

⁴Department of Respiratory Medicine, Luwan Branch, Ruijin Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, People's Republic of China

⁵Department of Pulmonary and Critical Care Medicine, Tongji Hospital, School of Medicine, Tongji University, No.389, Xincun Road, Shanghai 200065, China

⁶Department of Respiratory Medicine, Shanghai Ninth People's Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

⁷Department of Emergency, Ruijin Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai 200025, China

⁸Hangzhou Smart Intelligent Co., Ltd, Hangzhou, China

Received: 6 April 2025 / Accepted: 18 June 2025

Published online: 02 July 2025

References

1. Groff D, Sun A, Ssentongo AE, Ba DM, Parsons N, Poudel GR, et al. Short-term and Long-term rates of postacute sequelae of SARS-CoV-2 infection: A systematic review. *JAMA Netw Open*. 2021;4(10):e2128568.
2. Soriano JB, Murthy S, Marshall JC, Relan P, Diaz JV. A clinical case definition of post-COVID-19 condition by a Delphi consensus. *Lancet Infect Dis*. 2022;22(4):e102–7.
3. Liu Y, Gu X, Li H, Zhang H, Xu J. Mechanisms of long COVID: an updated review. *Chin Med J Pulmonary Crit Care Med*. 2023;1(4):231–40.
4. Subramanian A, Nirantharakumar K, Hughes S, Myles P, Williams T, Gokhale KM, et al. Symptoms and risk factors for long COVID in non-hospitalized adults. *Nat Med*. 2022;28(8):1706–14.
5. Thaweethai T, Jolley SE, Karlson EW, Levitan EB, Levy B, McCormsey GA, et al. Development of a definition of postacute sequelae of SARS-CoV-2 infection. *JAMA*. 2023;329(22):1934–46.
6. Han X, Fan Y, Alwalid O, Li N, Jia X, Yuan M, et al. Six-month Follow-up chest CT findings after severe COVID-19 pneumonia. *Radiology*. 2021;299(1):E177–86.

7. Hellemons ME, Huijts S, Bek LM, Berentschot JC, Nakshbandi G, Schurink CAM, et al. Persistent health problems beyond pulmonary recovery up to 6 months after hospitalization for COVID-19: A longitudinal study of respiratory, physical, and psychological outcomes. *Ann Am Thorac Soc*. 2022;19(4):551–61.
8. Latronico N, Peli E, Calza S, Rodella F, Novelli MP, Cella A, et al. Physical, cognitive and mental health outcomes in 1-year survivors of COVID-19-associated ARDS. *Thorax*. 2022;77(3):300–3.
9. Lambin P, Leijenaar RTH, Deist TM, Peerlings J, de Jong EEC, van Timmeren J, et al. Radiomics: the Bridge between medical imaging and personalized medicine. *Nat Rev Clin Oncol*. 2017;14(12):749–62.
10. Mayerhoefer ME, Materka A, Langs G, Häggström I, Szczypiński P, Gibbs P, et al. Introduction to radiomics. *J Nucl Med*. 2020;61(4):488–95.
11. Ather S, Kadir T, Gleeson F. Artificial intelligence and radiomics in pulmonary nodule management: current status and future applications. *Clin Radiol*. 2020;75(1):13–9.
12. Cruz JA, Wishart DS. Applications of machine learning in cancer prediction and prognosis. *Cancer Inf*. 2007;2:59–77.
13. Kourou K, Exarchos TP, Exarchos KP, Karamouzou MV, Fotiadis DI. Machine learning applications in cancer prognosis and prediction. *Comput Struct Biotechnol J*. 2015;13:8–17.
14. Nagpal P, Guo J, Shin KM, Lim JK, Kim KB, Comellas AP, et al. Quantitative CT imaging and advanced visualization methods: potential application in novel coronavirus disease 2019 (COVID-19) pneumonia. *BJR Open*. 2021;3(1):20200043.
15. Pan F, Ye T, Sun P, Gui S, Liang B, Li L, et al. Time course of lung changes at chest CT during recovery from coronavirus disease 2019 (COVID-19). *Radiology*. 2020;295(3):715–21.
16. Xu X, Yu C, Qu J, Zhang L, Jiang S, Huang D, et al. Imaging and clinical features of patients with 2019 novel coronavirus SARS-CoV-2. *Eur J Nucl Med Mol Imaging*. 2020;47(5):1275–80.
17. Huang G, Gong T, Wang G, Wang J, Guo X, Cai E, et al. Timely diagnosis and treatment shortens the time to resolution of coronavirus disease (COVID-19) pneumonia and lowers the highest and last CT scores from sequential chest CT. *AJR Am J Roentgenol*. 2020;215(2):367–73.
18. Zhao YM, Shang YM, Song WB, Li QQ, Xie H, Xu QF, et al. Follow-up study of the pulmonary function and related physiological characteristics of COVID-19 survivors three months after recovery. *EClinicalMedicine*. 2020;25:100463.
19. Chaganti S, Grenier P, Balachandran A, Chabin G, Cohen S, Flohr T, et al. Automated quantification of CT patterns associated with COVID-19 from chest CT. *Radiol Artif Intell*. 2020;2(4):e200048.
20. Chung M, Bernheim A, Mei X, Zhang N, Huang M, Zeng X, et al. CT imaging features of 2019 novel coronavirus (2019-nCoV). *Radiology*. 2020;295(1):202–7.
21. Bernheim A, Mei X, Huang M, Yang Y, Fayad ZA, Zhang N, et al. Chest CT findings in coronavirus Disease-19 (COVID-19): relationship to duration of infection. *Radiology*. 2020;295(3):200463.
22. China NHCPR. The diagnosis and treatment protocol for COVID-19 patient (the 10th Edition) 2023 [Available from: <https://data.who.int/dashboards/covid19/cases?n=c>]
23. Kim GB, Jung KH, Lee Y, Kim HJ, Kim N, Jun S, et al. Comparison of shallow and deep learning methods on classifying the regional pattern of diffuse lung disease. *J Digit Imaging*. 2018;31(4):415–24.
24. Park B, Park H, Lee SM, Seo JB, Kim N. Lung segmentation on HRCT and volumetric CT for diffuse interstitial lung disease using deep convolutional neural networks. *J Digit Imaging*. 2019;32(6):1019–26.
25. Hwang HJ, Seo JB, Lee SM, Kim EY, Park B, Bae HJ, et al. Content-Based image retrieval of chest CT with convolutional neural network for diffuse interstitial lung disease: performance assessment in three major idiopathic interstitial pneumonias. *Korean J Radiol*. 2021;22(2):281–90.
26. Kim MS, Choe J, Hwang HJ, Lee SM, Yun J, Kim N, et al. Interstitial lung abnormalities (ILA) on routine chest CT: comparison of radiologists' visual evaluation and automated quantification. *Eur J Radiol*. 2022;157:110564.
27. Podolanczuk AJ, Oelsner EC, Barr RG, Hoffman EA, Armstrong HF, Austin JH, et al. High Attenuation areas on chest computed tomography in community-dwelling adults: the MESA study. *Eur Respir J*. 2016;48(5):1442–52.
28. Sack CS, Doney BC, Podolanczuk AJ, Hooper LG, Seixas NS, Hoffman EA, et al. Occupational exposures and subclinical interstitial lung disease. The MESA (Multi-Ethnic study of Atherosclerosis) air and lung studies. *Am J Respir Crit Care Med*. 2017;196(8):1031–9.
29. Kemerink GJ, Lamers RJ, Pellis BJ, Kruize HH, van Engelshoven JM. On segmentation of lung parenchyma in quantitative computed tomography of the lung. *Med Phys*. 1998;25(12):2432–9.
30. Hatabu H, Hunninghake GM, Richeldi L, Brown KK, Wells AU, Remy-Jardin M, et al. Interstitial lung abnormalities detected incidentally on CT: a position paper from the Fleischner society. *Lancet Respir Med*. 2020;8(7):726–37.
31. Elliott MR, O'Connor AE, Marshall GD. Inflammatory pathways in patients with post-acute sequelae of COVID-19: the role of the clinical immunologist. *Ann Allergy Asthma Immunol*. 2024;133(5):507–15.
32. Peluso MJ, Deeks SG. Mechanisms of long COVID and the path toward therapeutics. *Cell*. 2024;187(20):5500–29.
33. Patrucco F, Solidoro P, Gavelli F, Apostolo D, Bellan M. Idiopathic pulmonary fibrosis and Post-COVID-19 lung fibrosis: links and risks. *Microorganisms*. 2023;11(4).
34. Ali RMM, Ghonimy MBI. Post-COVID-19 pneumonia lung fibrosis: a worrisome sequelae in surviving patients. *Egypt J Radiol Nuclear Med*. 2021;52(1):101.
35. Hama Amin BJ, Kakamad FH, Ahmed GS, Ahmed SF, Abdulla BA, Mohammed SH, et al. Post COVID-19 pulmonary fibrosis; a meta-analysis study. *Ann Med Surg* (Lond). 2022;77:103590.
36. Babar M, Jamil H, Mehta N, Moutwakil A, Duong TQ. Short- and Long-Term Chest-CT findings after recovery from COVID-19: A systematic review and Meta-Analysis. *Diagnostics* (Basel). 2024;14(6).
37. Alliou S, Zangiabadian M, Pouramini A, Jaberinezhad M, Shobeiri P, Ghozy S, et al. Radiological findings as predictors of COVID-19 lung sequelae: A systematic review and Meta-analysis. *Acad Radiol*. 2023;30(12):3076–85.
38. Caruso D, Guido G, Zerunian M, Polidori T, Lucertini E, Pucciarelli F, et al. Post-Acute sequelae of COVID-19 pneumonia: Six-month chest CT Follow-up. *Radiology*. 2021;301(2):E396–405.
39. Hu ZJ, Xu J, Yin JM, Li L, Hou W, Zhang LL, et al. Lower Circulating Interferon-Gamma is a risk factor for lung fibrosis in COVID-19 patients. *Front Immunol*. 2020;11:585647.
40. Li X, Shen C, Wang L, Majumder S, Zhang D, Deen MJ, et al. Pulmonary fibrosis and its related factors in discharged patients with new Corona virus pneumonia: a cohort study. *Respir Res*. 2021;22(1):203.
41. Johnston J, Dorrian D, Linden D, Stanel SC, Rivera-Ortega P, Chaudhuri N. Pulmonary sequelae of COVID-19: focus on interstitial lung disease. *Cells*. 2023;12(18).
42. Liu M, Lv F, Huang Y, Xiao K. Follow-Up study of the chest CT characteristics of COVID-19 survivors seven months after recovery. *Front Med (Lausanne)*. 2021;8:636298.
43. Roig-Martí C, Pérez-Catalán I, Varea-Villanueva M, Folgado-Escudero S, Navarro-Ballester A, Fernández-García MP, et al. Predictors of the presence of radiological abnormalities 6 months after severe COVID-19 pneumonia. *BMC Infect Dis*. 2024;24(1):883.
44. Najjar-Debbiny R, Barnett-Griness O, Khoury J, Gronich N, Weber G, Adir Y, et al. Association between COVID-19 infection and pulmonary fibrosis: A nested Case-Control study. *Am J Med*. 2023;136(11):1087–e932.
45. George PM, Wells AU, Jenkins RG. Pulmonary fibrosis and COVID-19: the potential role for antifibrotic therapy. *Lancet Respir Med*. 2020;8(8):807–15.
46. George PM, Reed A, Desai SR, Devaraj A, Faiez TS, Laverty S, et al. A persistent neutrophil-associated immune signature characterizes post-COVID-19 pulmonary sequelae. *Sci Transl Med*. 2022;14(671):eabo5795.
47. Radermecker C, Detrembleur N, Guiot J, Cavalier E, Henket M, d'Emal C et al. Neutrophil extracellular traps infiltrate the lung airway, interstitial, and vascular compartments in severe COVID-19. *J Exp Med*. 2020;217(12).
48. Mrigupuri P, Goel N, Spalgais S, Nagaraja R, Sonal S, Yadav SR, et al. Uncontrolled diabetes mellitus predicts Post-COVID-19 fibrosis. *J Assoc Physicians India*. 2023;71(7):11–2.
49. Salonen J, Lampela H, Kesitalo E, Vähäniikkilä H, Purokivi M, Kaarteenaho R. Bronchoalveolar lavage differential cell count on prognostic assessment of patients with stable or acute interstitial lung disease: A retrospective real-life study. *Clin Immunol*. 2020;220:108594.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.